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324751 (24)

BE (7th Semester)

Examination, April-May, 2013

Branch : Electrical, EEE

POWER APPARATUS SYSTEM

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question is compulsory.

(ii) Attempt any two parts from (b), (c) and (d) of each question.

Q. 1. (a) What are the factors affecting sag? 2

(b) Name different types of insulator used in power system. Describe the main construction of pin and string insulator. 7

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(c) An overhead line has ACSR conductor of 1.95 cm diameter and span of 244 m. The allowable tension is 3.56×10^4 N. Find : 7

(i) Sag in still air condition with no ice covering.

(ii) Vertical sag when there is an ice covering of 0.96 cm thick and horizontal wind pressure of 382 N/m^2 of projected area. Ice weighs 8920 N/m^3.

(iii) The line is carried by insulated string 1.43 m long. What should be height of lowest cross arm to give a minimum ground clearance of 7.62 m under bad weather conditions?

Conductor weight is 0.847 kg/m.

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(d) What is sag template? Explain vibration of conductor. 7

Q. 2. (a) What are touch and step voltages? 2

(b) How sub stations are classified on basis of utility and on basis of construction? 7

(c) Derive expression for earth resistance offered to hemispherical electrode. 7

(d) What are various steps involved in designing a sub-station? 7

Q. 3. (a) What is resistance earthing? 2

(b) Explain briefly : 7

(i) Arching ground

(ii) Reactance earthing

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(c) Derive expression for reactance of Peterson coil in terms of capacitance of protected line. A 132 KV, 3-ϕ, 50 Hz transmission line 80 km long has capacitance to earth of each line equal to 0.00914 μF/km. Determine inductance and KVA rating of arc supression coil suitable for this line. 7

(d) Discuss resistance grounding with neat phasor diagram. 7

Q. 4. (a) Define BIL. 2

(b) Describe courses of over-voltage in power system. 7

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(c) Describe mechanism and characteristic of lightning discharge. 7

(d) What is insulation coordination? How coordination among sub-station equipments done? 7

Q. 5. (a) A system formed by 3 components A, B, C in series. The reliabilities are 0.8, 0.86 and 0.92 respectively. Find reliability of system. 2

(b) Draw typical bath tub curve. Explain 3 regions of curve. For which region are reliability calculations valid? Why? 7

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(c) What is reliability planning? Explain factors required for preparation of reliability models? 7

(d) (i) Prove that $\lambda(t) = \frac{F(t)}{R(t)}$

Where,

$\lambda(t)$ = failure rate of system

$F(t)$ = probability density function

$R(t)$ = reliability of system 7

(ii) A radial system consists of transformer and distributor 0.4 km long. The failure rates for

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transformer and distributor are 0.8 failures per year and 8.2 failures per km per year respectively. Their repair times are 6 hr and 5 hr respectively. Find :

(1) Failure rate of system.

(2) Down time per outage.

(3) Total outage time per year. 7

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